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CASE STUDY / SURVEY REPORT

Subject: Embedded System (7IT04)

CASE STUDY TITLE

Architectural Analysis of an IoT-Based Embedded Smart Parking System

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INDEX / TABLE OF CONTENTS

Architectural Analysis of an IoT-Based Embedded Smart Parking System

SEC.	CHAPTER / SECTION TITLE	REPORT PG.	MERGED PG.
1.	**Abstract and Key Terms**	**1**	**4**
2.	**Introduction and Objective**	**1**	**4**
2.1	Context	1	4
2.2	Engineering Objectives and Measurable Targets	2	5
3.	**System Architecture / Block Diagram**	**3**	**6**
3.1	End-to-End Signal Flow	3	6
3.2	Layer Breakdown	3	6
4.	**Hardware Components and Technical Specifications**	**4**	**7**
4.1	Microcontroller Comparison: ESP32 vs STM32	4	7
4.2	Power Management	5	8
4.2.1	Power Tree	5	8
4.2.2	Average Power with Sleep Cycles	5	8
4.2.3	ESP32 Slot-Node Alternative (for comparison)	6	9
4.3	Displays and Indicators	6	9
4.4	Gate Access: Servo and Stepper Drivers	6	9
4.5	Pin Assignment: ESP32 Gate Controller	7	10
5.	**Software Components and Embedded Code Structure**	**8**	**11**
5.1	Architecture Choice: RTOS Tasks vs Event-Driven Super-Loop	8	11
5.2	Task Layout (Gate Controller)	8	11
5.3	Sensor Polling and Debouncing (HC-SR04)	9	12
5.4	Interrupt-Driven Entry/Exit Barrier Trigger	9	12
5.5	Wi-Fi / Cellular Reconnect Handler with SPIFFS Store-and-Forward	10	13
6.	**Working Principle**	**11**	**14**
6.1	End-to-End Walk-Through	11	14
6.2	Slot Allocation Algorithm	12	15
6.3	Latency Budget	12	15

INDEX / TABLE OF CONTENTS (CONTINUED)

SEC.	CHAPTER / SECTION TITLE	REPORT PG.	MERGED PG.
7.	**Communication Protocols Used**	**13**	**16**
7.1	On-Board Buses	13	16
7.2	Wireless Technology Comparison	14	17
7.3	MQTT vs HTTP/REST	14	17
7.4	Power-Budget Summary	15	18
8.	**Sensors and Actuators**	**16**	**19**
8.1	Sensor Comparison	16	19
8.2	Actuators	17	20
9.	**Advantages and Limitations**	**17**	**20**
9.1	Technical Advantages	17	20
9.2	Engineering Limitations	18	21
10.	**Real-World Applications**	**18**	**21**
11.	**Challenges and Possible Improvements**	**19**	**22**
11.1	Operational Challenges	19	22
11.2	Future Enhancements	19	22
12.	**Conclusion and References**	**20**	**23**
12.1	Conclusion	20	23
12.2	References ([1] to [16])	21	24

LIST OF FIGURES AND TECHNICAL TABLES

REF.	FIGURE / TECHNICAL TABLE DESCRIPTION	REPORT PG.	MERGED PG.
Fig. 1	**Four-layer block diagram of the smart parking system (Sec. 3.1)**	**3**	**6**
Tab. 2.2	Engineering Objectives and Quantitative Targets	2	5
Tab. 3.2	Four-Layer Architecture Breakdown (Perception, Edge, Comm., Cloud)	3–4	6–7
Tab. 4.1	Microcontroller Comparison: ESP32-WROOM-32E vs STM32L432KC	4–5	7–8
Tab. 4.3	Displays and Indicators Interface & Current Specifications	6	9
Tab. 4.4	Gate Access Actuators and Drivers (SG90, L298N, Relay Module)	6–7	9–10
Tab. 4.5	Pin Assignment: ESP32 Gate Controller GPIO Mapping	7	10
Tab. 5.1	Software Architecture: FreeRTOS Tasks vs Interrupt + Super-Loop	8	11
Tab. 5.2	FreeRTOS Task Layout, Priority, Stack, and Core Pinning	8	11
Tab. 6.3	End-to-End System Latency Budget	12–13	15–16
Tab. 7.2	Wireless Technology Comparison (Wi-Fi, BLE 5, LoRa, LTE-M/NB-IoT)	14	17
Tab. 7.3	Application Protocol Comparison: MQTT 3.1.1/5 vs HTTP/1.1 REST	14–15	17–18
Tab. 7.4	Daily Uplink Power-Budget Summary (LoRa SF7/SF12 vs Wi-Fi)	15	18
Tab. 8.1	Sensor Comparison (HC-SR04, IR, Magnetometer, Optical/ANPR)	16	19
Tab. 8.2	Actuator Specifications and Drive Circuits	17	20
Tab. 9.2	Engineering Limitations, Mechanisms, and Mitigations	18	21

Architectural Analysis of an IoT-Based Embedded Smart Parking System

1. Abstract and Key Terms

Abstract. Urban parking search contributes a measurable share of city-centre traffic and its associated emissions. This report analyses a three-tier, IoT-based embedded smart parking system comprising battery-powered slot-level sensing nodes, a mains-powered ESP32 gate controller and gateway, and a cloud back end that exposes occupancy and billing services to mobile and web clients. The perception layer fuses a magnetometer with ultrasonic ranging to reject single-sensor false positives. STM32L4-class slot nodes spend more than 99.7 % of their time in Stop 2 mode, giving an analytically estimated average current of about 60 µA and a battery life of roughly 4.5 years on two lithium AA cells. The gate controller runs FreeRTOS on a dual-core ESP32 and uses interrupt-to-task notification for deterministic barrier actuation (design target: ISR-to-actuation latency of 10 ms or less), while Wi-Fi/MQTT handling is isolated on a separate task with an SPIFFS-backed store-and-forward queue that tolerates network outages. LoRa carries slot-to-gateway telemetry and MQTT over TLS 1.2 carries gateway-to-cloud traffic, trading a quantified protocol overhead for confidentiality and integrity. Comparative analysis of sensors, protocols and power architectures shows that a hybrid design (magnetometer-primary, ultrasonicsecondary, LoRa uplink) gives the best balance of accuracy, cost and lifetime. The report closes with limitations, security considerations and a roadmap towards TinyML edge vision and solar energy harvesting.

Key Terms: Embedded IoT; FreeRTOS real-time scheduling; LoRa / MQTT; ultra-low-power design; multisensor fusion.

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2. Introduction and Objective

2.1 Context

- **Congestion.** Studies of urban cruising for parking commonly attribute roughly 10 to 30 % of central-district traffic to drivers searching for a space (figures vary by city and study; cite the source used in your viva).
- **Emissions.** Idling and low-speed circulation have poor fuel efficiency per kilometre, so every minute of search adds CO2 and NOx. Guiding a driver directly to a free slot is a software-only emission-reduction measure once the sensing infrastructure exists.
- **Smart-city fit.** Parking occupancy is a low-bandwidth, high-value data stream: a few bytes per event, but it feeds navigation, dynamic pricing, enforcement and urban planning.

2.2 Engineering Objectives and Measurable Targets

#	Objective	Quantitative target	Design consequence

O1	Real-time occupancy sensing	State change detected and reported in 2 s or less; detection accuracy 98 % or better	Sensor fusion plus 3sample confirmation
O2	Automated billing / entry control	Barrier command latency of 10 ms or less after trigger; billing record created on every exit	ISR-driven gate path, idempotent billing events
O3	Cloud connectivity	99 % message delivery despite outages	QoS 1, SPIFFS storeand-forward
O4	Low power	Slot node life of 4 years or more on 2 x AA	Stop 2 duty cycling, LoRa, event-driven uplink
O5	Minimal latency	Slot change visible in app within 3 s	LoRa SF7, MQTT push, no polling by clients

The central engineering tension is that O1/O5 (fast, accurate, frequent sensing) directly oppose O4 (energy). Most architectural choices in this report are resolutions of that tension.

3. System Architecture / Block Diagram

3.1 End-to-End Signal Flow

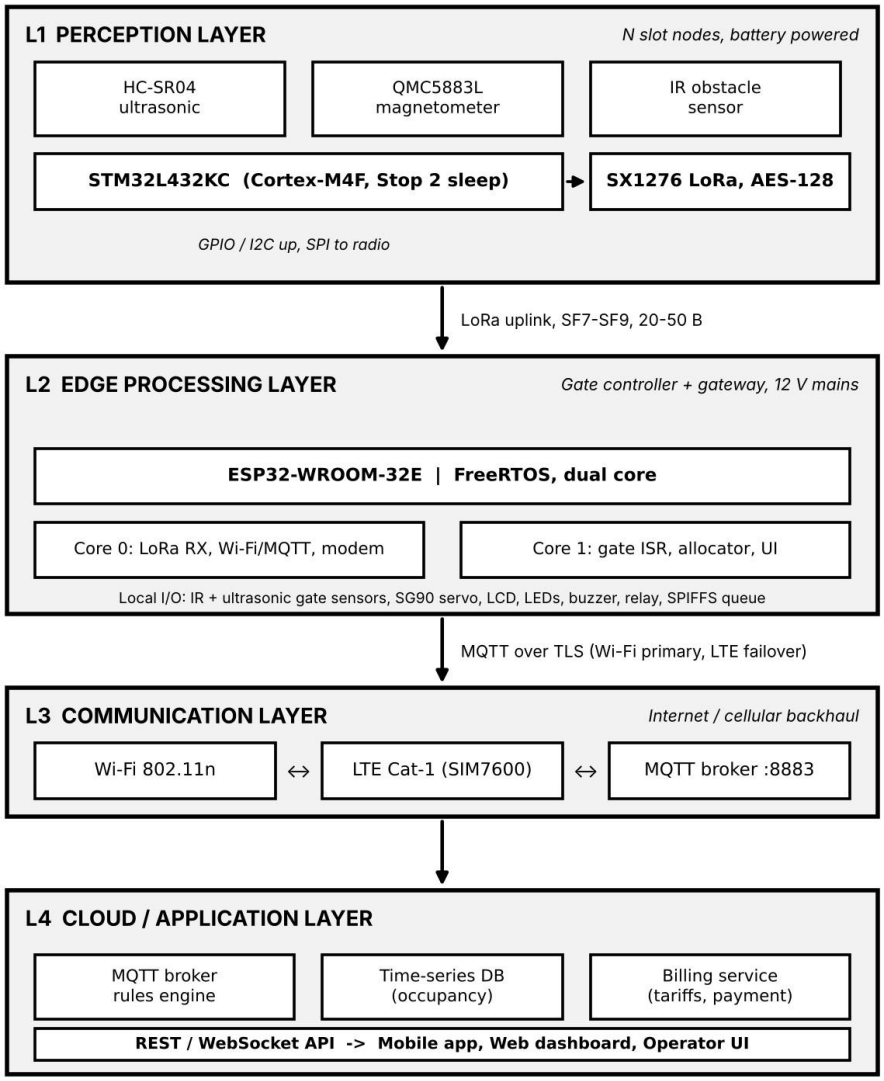


Figure 1: Four-layer block diagram of the smart parking system (black and white).

3.2 Layer Breakdown

Layer	Responsibility	Key hardware / software	Dominant design constraint
Perception	Convert physical presence into a debounced binary state plus confidence	HC-SR04, QMC5883L, IR, STM32L432	Energy per sample

Layer	Responsibility	Key hardware / software	Dominant design constraint
Edge processing	Real-time gate control, slot allocation, protocol translation, offline buffering	ESP32, FreeRTOS, SPIFFS	Determinism and fault tolerance
Communication	Move small, frequent messages reliably and securely	LoRa, Wi-Fi, LTE, MQTT/TLS	Range, airtime, power
Cloud / application	Persistence, billing, analytics, user interfaces	Broker, DB, REST API, mobile app	Scalability, security

A key architectural decision is **pushing decisions to the edge**: the gate must open even when the cloud is unreachable, so the authoritative real-time occupancy counter lives on the gate controller and the cloud holds an eventually consistent replica.

[1]. Hardware Components and Technical Specifications

Parameter	ESP32-WROOM-32E (gate/gateway)	STM32L432KC (slot node)
Core	Dual Xtensa LX6	Single Cortex-M4F with FPU
Max clock	240 MHz	80 MHz
Supply	3.0 to 3.6 V	1.71 to 3.6 V
Active, CPU only	about 30 to 70 mA (clock dependent)	about 84 µA/MHz, i.e. roughly 6.7 mA at 80 MHz
Wi-Fi TX peak	about 180 to 240 mA (brownout risk)	none (external radio)
Light / modem sleep	about 0.8 mA light sleep	Sleep, low-power run available
Deep sleep / Stop	about 10 µA (chip, RTC timer)	Stop 2 about 1 to 1.4 µA with RTC

[1] **.1 Microcontroller Comparison: ESP32 vs STM32**

Lowest state	Hibernation about 5 µA	Standby about 0.3 µA, Shutdown about 30 nA
ADC	12-bit SAR, non-linear, ADC2	12-bit, up to 16-bit by
Parameter	ESP32-WROOM-32E (gate/gateway)	STM32L432KC (slot node)
	unusable while Wi-Fi is on	oversampling, low-power capable
Wireless	Wi-Fi 802.11 b/g/n, BLE 4.2	none
Wake latency from deep sleep	tens of ms, plus Wi-Fi association time of 1 to 3 s	a few µs to tens of µs from Stop 2

Design decision. The ESP32 is chosen where connectivity and compute matter and mains power exists. The STM32L4 is chosen for slot nodes because its 1 µA-class Stop 2 mode and fast wake dominate battery life. An ESP32 slot node would be dominated by Wi-Fi association energy (Section 4.2.3).

4.2 Power Management

4.2.1 Power Tree

```
12 V DC adapter ---> MP1584/LM2596 buck ---> 5.0 V rail ---> servo, LCD backlight, HC-SR04
                                          |
                                          +-> buck 3.3 V (or LDO for low load) ---> ESP32, LoRa, IR

Slot node: 2 x Li-FeS2 AA (3.0 V nominal) ---> STM32L432 + SX1276 directly (no regulator,
           or low-Iq LDO such as TPS7A02, Iq about 25 nA)
Solar option: 6 V 1 W panel ---> CN3791 MPPT charger ---> 1 x 18650 ---> 3.3 V buckboost
```

LDO vs buck. An LDO dissipates P = (Vin - Vout) x I. For 5 V to 3.3 V at 240 mA: P = 1.7 V x 0.24 A = 0.41 W, giving 66 % efficiency and noticeable heating. A buck at 90 % efficiency delivers 0.79 W with a loss of about 0.09 W. LDOs remain preferable on the low-noise analogue rail of the magnetometer and where Iq (quiescent current) matters more than conversion loss.

Servo rail isolation. An SG90 draws 100 to 250 mA moving and up to roughly 650 mA at stall. Powering it from the logic rail causes brownouts and ESP32 resets. It is given its own 5 V rail with a 470 to 1000 µF bulk capacitor at the servo connector and a common ground star point.

4.2.2 Average Power with Sleep Cycles

```
I_avg = ( I_active x t_active + I_sleep x t_sleep ) / T,   T = t_active + t_sleep
P_avg = V_supply x I_avg
Duty cycle D = t_active / T
Battery life (h) = (C_nominal x eta_usable) / I_avg
```

Worked example: STM32 + HC-SR04 + LoRa slot node (3.0 V).

- Sampling period T = 10 s. Active window 30 ms at 18 mA (MCU 3 mA + HC-SR04 15 mA).
- Sleep: 9.97 s at 2 µA (Stop 2 + RTC + leakage).
- Charge per cycle: 0.030 s x 18 mA = 0.540 mC; sleep 9.97 s x 0.002 mA = 0.0199 mC; total 0.560 mC.
- I_avg(sensing) = 0.560 mC / 10 s = **56 µA**.
- LoRa heartbeat every 600 s: SF7, 20 B payload, about 56 ms airtime at 44 mA (+14 dBm) = 2.46 mC per 600 s = **4.1 µA**. State-change uplinks (about 40 per day) add roughly 0.1 mC x 40 / 86400 s, about 0.05 µA, which is negligible.
- Total I_avg of about **60 µA**, P_avg of about 0.18 mW.
- With C = 3000 mAh and 80 % usable: life = 2400 mAh / 0.060 mA = 40,000 h, about **4.6 years** (self-discharge and temperature derating make 4 years a safe claim).

Using a magnetometer as primary sensor (QMC5883L, about 100 µA in continuous mode, or single-shot measurement of about 1 ms) removes the 15 mA ultrasonic pulse from most cycles and lowers I_avg to roughly 10 to 15 µA, which is why the ultrasonic sensor is used as a secondary confirmation only.

4.2.3 ESP32 Slot-Node Alternative (for comparison)

Wake every 10 min, Wi-Fi associate and publish: about 1.5 s at 130 mA = 195 mC per 600 s, i.e. 0.325 mA, plus 10 µA sleep, giving I_avg of about 0.34 mA. Life = 2400 / 0.34 = 7,060 h, about **0.8 years**. This is roughly 6 x worse than the STM32 + LoRa node, and it cannot also sample every 10 s without waking the radio.

4.3 Displays and Indicators

Item	Interface	Voltage / current	Notes
16x2 LCD + PCF8574 backpack	I2C, address 0x27 or 0x3F	5 V, about 2 mA logic + 20 mA backlight	HD44780 needs 5 V; the PCF8574 opendrain lines are pulled up to 5 V, so use a bidirectional level shifter for 3.3 V I2C or power the backpack at 3.3 V and accept dimmer contrast
0.96 in SSD1306 OLED	I2C 0x3C (or SPI)	3.3 V, about 8 to 20 mA depending on pixels lit	Better contrast, no backlight, lower power, burn-in over years
Status LEDs (green free / red full)	GPIO	330 ohm series: (3.3 - 2.0 V) / 330 = about 4 mA	Use a transistor if more than 12 mA per pin

4.4 Gate Access: Servo and Stepper Drivers

Actuator / driver	Control	Electrical	Engineering comment
SG90 micro servo	50 Hz PWM, 1.0 to 2.0 ms pulse (0 to 180 degrees)	4.8 to 6 V, 100 to 250 mA moving, up to 650 mA stall, torque about 1.8 kg-cm	Adequate for a demonstrator barrier arm; use MG996R for real arms
Actuator / driver	Control	Electrical	Engineering comment
L298N + stepper / DC motor	2 x IN pins + ENABLE (PWM)	5 to 35 V, 2 A per channel	Bipolar-transistor output drops about 1.4 to 2 V, wasting power and heating. Acceptable for teaching; prefer DRV8833 or TB6612FNG (MOSFET, lower drop)
Relay module (barrier controller dry contact)	GPIO through optocoupler	5 V coil, about 70 mA	Add flyback diode on bare relays

Servo timing at 16-bit LEDC resolution and 50 Hz: duty = 65535 x (pulse_ms / 20 ms), so 1.5 ms gives 4915 counts.

4.5 Pin Assignment: ESP32 Gate Controller

Function	GPIO	Interface / note
LCD / OLED SDA, SCL	21, 22	I2C0, 100 kHz (400 kHz fast mode for OLED), 4.7 kohm pullups
LoRa SCK, MISO, MOSI, NSS	18, 19, 23, 5	VSPI, 8 MHz
LoRa DIO0 (RX done), RESET	14, 4	DIO0 as rising-edge interrupt
Modem UART2 RX, TX	16, 17	115200 8N1 to SIM7600
Ultrasonic TRIG, ECHO	25, 34	ECHO via 1 k / 2 k divider (5 V to 3.3 V); GPIO34 is input-only
IR entry, IR exit	35, 32	Digital, active low; GPIO35 input-only
Servo PWM	13	LEDC channel 0, 50 Hz

Buzzer, relay	33, 26	Driven through NPN / optocoupler
LED green, red	27, 15	GPIO15 is a boot-strapping pin (harmless LED load)

GPIO 6 to 11 (flash), 0, 2 and 12 (strapping pins) are deliberately avoided, and ADC2 pins are not used for analogue sensing because ADC2 is unavailable while Wi-Fi is active.

5. Software Components and Embedded Code Structure

5.1 Architecture Choice: RTOS Tasks vs Event-Driven Super-Loop

Criterion	FreeRTOS tasks (gate/gateway)	Interrupt + super-loop (slot node)
Concurrency	Preemptive, priority-based, separate Wi-Fi and control paths	One context plus ISRs; blocking calls stall everything
Determinism	Bounded by priority and tick (1 ms)	Excellent for a single task, poor as features grow
RAM / flash overhead	Tens of kB; each task needs its own stack	Minimal
Power	Tickless idle is possible but needs care with Wi-Fi	Natural: ISR wakes MCU, loop runs, return to Stop 2
Verification	Needs analysis of priority inversion and stack sizing	Simple state machine

Decision. The gate/gateway uses FreeRTOS because Wi-Fi/TLS stalls must never delay the barrier. The slot node uses an event-driven loop (RTC wake -> sample -> decide -> transmit -> Stop 2) because it has one job and a 2 µA sleep budget.

5.2 Task Layout (Gate Controller)

Task	Priority	Stack	Trigger	Core
gateTask	5 (highest)	3 kB	Task notification from ISR	1
sensorTask	4	3 kB	50 ms periodic (vTaskDelayUntil)	1

allocTask	3	4 kB	Queue of slot events	1
loraTask	3	4 kB	DIO0 interrupt	0
netTask (WiFi/MQTT)	2	8 kB (TLS)	Event + backoff timer	0
uiTask (LCD/LEDs)	1	2 kB	250 ms periodic	1

Shared state (free-slot count) is protected by a mutex with priority inheritance, or modified only by allocTask, with other tasks sending messages to it. The ISR never calls blocking APIs.

5.3 Sensor Polling and Debouncing (HC-SR04)

An N-of-M confirmation filter with hysteresis avoids chattering when a vehicle edge sits near the threshold.

```
#include <Arduino.h>
constexpr uint8_t TRIG = 25, ECHO = 34;
constexpr float OCC_CM = 40.0f, FREE_CM = 60.0f;   // hysteresis band
constexpr uint8_t CONFIRM = 3;                      // consecutive agreeing samples

inline float soundMs(float tC) { return 331.3f + 0.606f * tC; }  // m/s

float readDistanceCm(float tC) {
  digitalWrite(TRIG, LOW);   delayMicroseconds(2);
digitalWrite(TRIG,  HIGH);  delayMicroseconds(10);
digitalWrite(TRIG, LOW);
  uint32_t us = pulseIn(ECHO, HIGH, 25000UL);       // 25 ms timeout = about 4.3
m   if (us == 0) return NAN;                          // no echo   return (us *
1e-6f * soundMs(tC) / 2.0f) * 100.0f; // d = t*v/2 }  bool occupied = false;
uint8_t agree = 0;

void sensorTask(void*) {
  TickType_t last = xTaskGetTickCount();
for (;;) {
    float d = readDistanceCm(readTempC());          // temperature-compensated
bool wantOcc = occupied;     if (!isnan(d)) {
      if (d < OCC_CM)  wantOcc = true;
      if (d > FREE_CM) wantOcc = false;             // between thresholds: hold state
    }
    if (wantOcc != occupied) {
      if (++agree >= CONFIRM) { occupied = wantOcc; agree = 0;
publishSlotEvent(occupied); }
    } else agree = 0;
    vTaskDelayUntil(&last, pdMS_TO_TICKS(50));      // fixed period, no drift
}
}
```

At 50 ms polling and 3-sample confirmation, worst-case detection delay is about 150 ms plus ranging time, well inside the 2 s target. `pulseIn` blocks for up to 25 ms; production code should use the ECHO edge capture on a hardware timer (input capture) to avoid CPU blocking.

5.4 Interrupt-Driven Entry/Exit Barrier Trigger

The ISR does the minimum: timestamp-based debounce and a notification to the high-priority task.

```
static TaskHandle_t gateTaskH;
constexpr uint32_t DEBOUNCE_US = 50000;
 void IRAM_ATTR isrEntry() {
static uint32_t lastUs = 0;
uint32_t now = micros();
  if (now - lastUs < DEBOUNCE_US) return;           // reject contact/optical chatter

  lastUs = now;
  BaseType_t hp = pdFALSE;
  vTaskNotifyGiveFromISR(gateTaskH, &hp);
  portYIELD_FROM_ISR(hp);                           // context switch immediately if
needed
}
void gateTask(void*) {
for (;;) {
    ulTaskNotifyTake(pdTRUE, portMAX_DELAY);        // sleep until ISR notifies     if
(freeSlots() > 0 && vehicleConfirmed()) {    // ultrasonic cross-check       uint32_t
ticket = createSession();            // entry timestamp, plate or RFID id
      setServoUs(2000);                             // open (about 90 degrees)
while (beamBlocked() && !timeout(15000)) vTaskDelay(pdMS_TO_TICKS(20)); // safety
      vTaskDelay(pdMS_TO_TICKS(1500));
      setServoUs(1000);                             // close
      enqueueEvent(EV_ENTRY, ticket);               // handled by netTask
    } else {
      signalFull();                                 // red LED, buzzer, LCD: FULL
}
  }
}
void setup() {
  xTaskCreatePinnedToCore(gateTask, "gate", 3072, nullptr, 5, &gateTaskH,
1);   attachInterrupt(digitalPinToInterrupt(35), isrEntry, FALLING); }
```

Safety logic: the barrier must never close while the IR safety beam is blocked, and a watchdog (esp_task_wdt) resets the controller if gateTask hangs with the arm lowered onto a vehicle.

5.5 Wi-Fi / Cellular Reconnect Handler with SPIFFS Store-and-Forward

Offline events are appended to a flash file and replayed in order once the link returns (at-least-once delivery, so the cloud de-duplicates on a unique event id).

```
bool storeEvent(const String& json) {               // called when offline
  File f = SPIFFS.open("/queue.log", FILE_APPEND);
if (!f) return false;
  if (f.size() > 64 * 1024) { f.close(); dropOldest(); f = SPIFFS.open("/queue.log",
FILE_APPEND); }
  f.println(json); f.close();
return true;
}
void flushQueue() {                                 // called after reconnect
  File f = SPIFFS.open("/queue.log", FILE_READ);
File tmp = SPIFFS.open("/queue.tmp", FILE_WRITE);
while (f.available()) {
    String line = f.readStringUntil(10);
    if (!mqtt.publish("parking/lot1/events", line.c_str(), false)) tmp.println(line);
// keep unsent
else mqtt.loop();

  }
  f.close(); tmp.close();
  SPIFFS.remove("/queue.log"); SPIFFS.rename("/queue.tmp", "/queue.log"); }
void netTask(void*) {   uint32_t
backoffMs = 1000;   for (;;) {
    if (WiFi.status() != WL_CONNECTED) WiFi.reconnect();
    if (WiFi.status() != WL_CONNECTED && modemAttach()) useCellular();  // LTE failover
    if (linkUp() && !mqtt.connected() && mqtt.connect(clientId, user, pass)) {
mqtt.subscribe("parking/lot1/cmd", 1);       flushQueue();       backoffMs =
1000;     } else if (!linkUp()) {
      backoffMs = min(backoffMs * 2, 60000u);       // exponential backoff, capped
    }
Event e;
    while (xQueueReceive(evQ, &e, 0)) {
String j = toJson(e);
      if  (!(mqtt.connected()  &&  mqtt.publish("parking/lot1/events",  j.c_str())))
storeEvent(j);
    }
    mqtt.loop();
    vTaskDelay(pdMS_TO_TICKS(mqtt.connected() ? 100 : backoffMs));   }
}
```

Design notes: (1) exponential backoff avoids hammering a failed access point and saves energy; (2) the ring cap stops flash exhaustion; (3) SPIFFS has wear levelling, but at one event per few seconds the lifetime (around 100,000 erase cycles per sector) is not a concern; (4) the 8 kB task stack accommodates mbedTLS handshake buffers.

6. Working Principle

6.1 End-to-End Walk-Through

1. **Vehicle arrival.** A car approaches the entry lane. The entry IR beam (GPIO35) is interrupted, producing a falling edge and firing `isrEntry()`. The ISR debounces (50 ms), then notifies `gateTask`.
2. **Verification.** `gateTask` wakes and takes a single HC-SR04 reading to confirm a vehicle-sized object at 30 to 150 cm. This ultrasonic cross-check rejects pets, bicycles and falling leaves that break the IR beam (see Section 11.1). If the reading is invalid, the event is discarded.
3. **Slot allocation.** If the free-slot counter is greater than zero, the allocator chooses a slot (6.2). The LCD shows the slot id; the zone LED turns green.
4. **Gate actuation.** The servo is driven to 2000 µs (open). The arm stays open until the safety beam clears, then closes after a 1.5 s dwell.
5. **Slot-level confirmation.** The slot node detects the vehicle (magnetometer shift plus ultrasonic range below 40 cm, confirmed over 3 samples) and sends a LoRa frame (slot id, state, battery, RSSI-independent sequence number). The gateway reconciles: expected slot occupied equals confirmed.
6. **Cloud payload.** The gateway publishes a JSON event over MQTT QoS 1, for example `{id:'e-8841', t:1760350000, ev:'ENTRY', slot:'B07', plate:null}`. If the link is down it is queued in SPIFFS (Section 5.5).
7. **Exit processing.** At exit, the same IR/ultrasonic chain fires. The controller closes the session: duration = t_exit - t_entry.
8. **Auto-billing.** The cloud computes the fee and triggers payment. The barrier opens when payment is confirmed (online) or when a pre-paid or grace-period token is valid (offline mode).

```
fee = rate x ceil( max(0, (t_exit - t_entry) - t_grace) / dt ),  capped at daily
maximum
Example: rate = 20 INR per hour, dt = 1 h, t_grace = 15 min, stay = 2 h 40 min
fee = 20 x ceil( (160 - 15) / 60 ) = 20 x 3 = 60 INR
```

6.2 Slot Allocation Algorithm

Each free slot s is given a cost; the minimum is assigned:

```
cost(s) = w1 * d_path(entry, s) + w2 * P_reserved(s) + w3 * P_zone(s) + w4 * stale(s)

d_path   : walking or driving distance from the entry (Dijkstra on the lot graph,
precomputed once, stored as a sorted table in flash)
P_reserved: large penalty if the slot is reserved for a different user class (EV,
disabled)
P_zone   : balances load across levels to avoid congesting one ramp
stale(s) : penalty if the slot node has not reported for more than 2 heartbeats
(suspect)
```

Because the table is precomputed, runtime allocation is O(n) over free slots (n of a few hundred), executing in well under 1 ms on the ESP32.

6.3 Latency Budget

Stage	Typical latency	Source
IR edge to ISR	under 10 µs	Interrupt latency
ISR debounce and notify to gateTask running	about 50 to 200 µs	FreeRTOS context switch
Ultrasonic verification	1 to 25 ms	Echo time-of-flight and timeout
Servo travel (SG90, about 0.1 s per 60 degrees)	150 to 300 ms	Mechanical
Slot node sample, confirm (3 x 100 ms in active burst)	about 300 ms	Debounce policy
LoRa SF7 airtime	about 57 ms	Section 7.4
Gateway to broker (TLS session	30 to 150 ms	Network RTT
Stage	Typical latency	Source
already open)		
Broker to app push	50 to 200 ms	Cloud

Electronics reaction to gate command (ISR to servo PWM change) is about 1 ms, comfortably meeting the 10 ms target; end-to-end slot-change visibility is about 1 s, meeting the 3 s target.

7. Communication Protocols Used

7.1 On-Board Buses

I2C (LCD, OLED, magnetometer). Open-drain, two wires, address-based. Timing: 100 kHz (standard) or 400 kHz (fast), data valid while SCL is high.

```
SCL  ____|^|_|^|_|^|_|^|_|^|_|^|_|^|_|^|_|^|_|^|____|^|___
SDA  ^^^^^|___A6__A5__A4__A3__A2__A1__A0__R/W__ACK___ ... STOP: SDA rises while SCL
high
          START: SDA falls while SCL high
```

Pull-up sizing from the I2C specification (rise time t_r, bus capacitance C_b, 0.8473 factor for 30 to 70 %):

```
R_p(max) = t_r / (0.8473 x C_b)       100 kHz: t_r = 1000 ns, C_b = 100 pF -> R_p <=
11.8 kohm
                                       400 kHz: t_r = 300 ns,  C_b = 100 pF -> R_p <=
3.5 kohm
R_p(min) = (V_dd - V_OL) / I_OL       (3.3 - 0.4) V / 3 mA = 967 ohm
```

A practical value is 2.2 to 4.7 kohm. **Bus contention management:** unique addresses (LCD 0x27, SSD1306 0x3C, QMC5883L 0x0D); a FreeRTOS mutex around the Wire bus because two tasks must not interleave transactions; hardware arbitration handles multi-master collisions but is not used here; clock stretching by slow slaves is tolerated by setting a timeout, and a bus-recovery routine (9 SCL pulses) frees a slave holding SDA low after a brownout.

SPI (SX1276 LoRa, optional SD card). Full duplex, 4 wires, Mode 0 (CPOL 0, CPHA 0) for the SX1276 at up to 10 MHz, here 8 MHz. Contention is avoided by one chip-select per device, MISO tri-stated by the device when its NSS is high, and a mutex (`xSemaphoreTake(spiMtx)`) so only one task owns the bus. A shared SD card on the same bus is the usual cause of corrupted LoRa transfers when CS lines are not idled high at boot.

UART (SIM7600 LTE modem, GPS, debug). Asynchronous 8N1. At 115200 baud the bit time is 1/115200 = 8.68 µs and a 10-bit frame takes 86.8 µs (about 11.5 kB/s). Receiver error budget is roughly plus or minus 3 %, so clock mismatch must be low (use a crystal-derived baud generator, not an RC oscillator). Hardware flow control (RTS/CTS) is enabled for the modem to avoid FIFO overrun during PPP or ATcommand bursts.

```
Idle=1 | START=0 | D0 D1 D2 D3 D4 D5 D6 D7 (LSB first) | STOP=1 | Idle
         <-- 8.68 us per bit at 115200 baud -->
```

7.2 Wireless Technology Comparison

Metric	Wi-Fi 802.11n	BLE 5	LoRa / LoRaWAN	LTE-M / NB-IoT
Typical range	30 to 100 m	10 to 100 m (LE Coded longer)	2 to 5 km urban, 10+ km rural	Cell coverage
Data rate	up to 72 Mb/s	125 kb/s to 2 Mb/s	0.3 to 5.5 kb/s (SF12 to SF7), 125 kHz	about 100 kb/s to 1 Mb/s
Latency	5 to 50 ms	3 to 30 ms	60 ms to 1.3 s airtime per frame	100 ms to several s
TX current	130 to 240 mA	5 to 15 mA	20 to 125 mA	100 to 300 mA
Join/association cost	1 to 3 s at 100+ mA	tens of ms	none (ALOHA)	several s
Energy per 20 B message	about 150 to 400 mJ with association; about 2 mJ if the connection persists	about 0.1 mJ	about 0.15 mJ (SF7), about 4 mJ (SF12)	tens of mJ

Best use here	Gateway uplink (mains)	Commissioning / maintenance	Slot-to-gateway	Backhaul failover

LoRa time-on-air (Semtech):

```
T_sym     = 2^SF / BW
T_preamble= (n_pre + 4.25) x T_sym
n_payload = 8 + max( ceil( (8 PL - 4 SF + 28 + 16 CRC - 20 H) / (4 (SF - 2 DE)) ) x
(CR + 4), 0 )
ToA       = T_preamble + n_payload x T_sym

SF7,  BW125, PL=20 B, CR4/5, CRC on: T_sym=1.024 ms, n_payload=43 -> ToA = 56.6 ms
SF9:                                 T_sym=4.096 ms, n_payload=33 -> ToA about 185 ms
SF12 (DE=1):                         T_sym=32.8 ms,  n_payload=28 -> ToA about 1.32 s
```

SF12 uses roughly 23 x the airtime and energy of SF7, so adaptive data rate (ADR) is essential. Where sub-GHz duty-cycle limits apply (EU868: 1 % per sub-band), a node using SF9 at 185 ms may transmit at most 1 frame per 18.5 s; verify the regulation for the deployment country (India: 865 to 867 MHz band rules).

7.3 MQTT vs HTTP/REST

Aspect	MQTT 3.1.1 / 5 over TCP	HTTP/1.1 REST over TCP
Pattern	Publish/subscribe, brokermediated	Request/response, client polls
Aspect	MQTT 3.1.1 / 5 over TCP	HTTP/1.1 REST over TCP
Fixed header	2 bytes plus topic	Hundreds of bytes of headers per request
Connection	One persistent, keep-alive (for example 60 s)	New connection per request unless kept alive
Server-to-device push	Native (subscribe)	Needs polling or WebSocket
Delivery guarantees	QoS 0, 1, 2; Last Will and Testament	None built in; app-level retries
Fit for constrained devices	Excellent	Acceptable for infrequent, large payloads

For a 60-byte payload, a REST POST with typical headers is about 400 to 800 bytes on the wire (plus TLS), whereas an MQTT publish is about 70 to 90 bytes. **TLS cost:** a full TLS 1.2 handshake costs about 4 to 6 kB of traffic, 2 round-trips, tens of kB of RAM and about 1 to 2 s of ESP32 CPU time for ECDHE/RSA operations; a persistent session or TLS session resumption amortises this once per reconnect. Last Will and Testament is used so the broker publishes `status: offline` if the gateway disappears, giving the operator instant fault visibility.

7.4 Power-Budget Summary

Link	Energy per 20 B uplink	Messages per day (assume 100)	Daily energy
LoRa SF7	about 8 mJ at 3.3 V (56.6 ms x 44 mA x 3.3 V)	100	0.8 J
LoRa SF12	about 190 mJ	100	19 J
Wi-Fi with association	about 0.65 J (1.5 s x 130 mA x 3.3 V)	100	65 J

This explains the choice of LoRa for battery nodes: at SF7 it is roughly 80 x cheaper than Wi-Fi with a cold association.

8. Sensors and Actuators 8.1

Sensor Comparison

Parameter	HC-SR04 ultrasonic	IR obstacle (reflective)	Ground magnetometer (AMR/QMC5883L-class)	Optical / ANPR camera
Principle	Time of flight, 40 kHz	Reflected IR intensity	Perturbation of Earth field by ferrous mass	Image processing / OCR
Range	2 to 400 cm, beam about 15 degrees	2 to 30 cm	Contact / buried, detects within about 1 m	3 to 15 m
Accuracy	About 3 mm ideal; 1 to 2 cm practical	Binary, threshold dependent	95 to 99 % detection (vendor claims)	90 to 99 % plate recognition, light dependent
Supply current	15 mA active	about 20 mA (emitter)	about 0.1 mA continuous, about 1 µA average duty-cycled	100 to 300 mA plus compute

Temperature	v varies 0.6 m/s per degree C (0.17 % per degree C)	Mild	Drift with temperature; baseline recalibration	Mild (thermal noise)
Rain / dust	Poor: droplets and dirt on the transducer, wind noise	Poor: lens fouling	Excellent (sealed, buried)	Rain, glare and dirt degrade OCR
Ambient light	Immune	Sunlight saturates the receiver	Immune	Night needs IR illumination
Line of sight	Required	Required	Not required	Required
Typical cost	USD 1 to 3	USD 0.5 to 2	USD 2 (module) to 30+ (sealed node)	USD 200 to 1000+ per lane

Ultrasonic mathematics.

```
Distance = (Time x Speed_of_sound) / 2
v(T)     = 331.3 + 0.606 x T   [m/s, T in degrees C]
Example: echo = 1160 us at 25 C -> v = 346.45 m/s -> d = 1160e-6 x 346.45 / 2 = 0.2009
m
```

```
Rule of thumb: about 58 us per cm of distance (round trip at 343 m/s)
Temperature error if 20 C is assumed but the air is 35 C: v = 352.5 vs 343.4 m/s, 2.6
% error, which is 5.3 cm at 2 m. Compensate with a DS18B20 or the MCU
temperature sensor.
```

Magnetometer detection. Earth's field is about 25 to 65 µT; a car roof or chassis perturbs it by tens to hundreds of µT at slot level.

```
B = sqrt(Bx^2 + By^2 + Bz^2);  dB = |B - B_base|
B_base updated by an exponential moving average only while the slot is free:
B_base(k) = (1 - alpha) B_base(k-1) + alpha B(k),   alpha about 0.01
Occupied if dB > threshold for N consecutive samples (hysteresis between on and off
thresholds).
```

The adaptive baseline tracks thermal drift, but adjacent-slot cars and passing traffic cause cross-talk, which is why the ultrasonic check is combined as a second vote (fusion rule: occupied only if magnetic AND ultrasonic agree; if they disagree for more than 3 cycles, flag the node for maintenance). **8.2 Actuators**

Actuator	Role	Drive	Notes
SG90 servo	Barrier arm (demonstrator)	3.3 V logic PWM, 5 V power	Detach PWM when idle to avoid jitter and save about 10 mA

Solenoid (12 V)	Bollard lock / slot lock	N-MOSFET (logic level) plus flyback diode, PWM hold current reduction	Peak 1 to 2 A at pull-in; hold at 30 % duty to limit heating
Piezo buzzer	Warnings (full lot, obstruction)	NPN transistor, 2 to 4 kHz tone via LEDC	Active vs passive: passive allows distinct tones
Relay module	Mains-rated barrier motor controller contact	Optocoupler, 5 V coil about 70 mA	Observe creepage; do not switch mains from the PCB without isolation

9. Advantages and Limitations

9.1 Technical Advantages

- **Automated space tracking.** Real-time occupancy replaces manual counting and removes human error; the gate counter is cross-checked by per-slot sensing, so drift (cars that tailgate or leave via wrong lane) is self-correcting.
- **Reduced congestion.** Guidance to a free slot removes the cruising component of traffic (Section 2.1).
- **Predictive slot allocation.** The time-series database yields arrival-rate and dwell-time distributions, enabling short-horizon occupancy forecasts (for example exponential smoothing or ARIMA) and dynamic pricing.
- **Resilience.** Local authority at the gate plus store-and-forward means the lot keeps operating during cloud outages.
- **Low operational cost.** Years of battery life and wireless installation avoid trenching cable. **9.2 Engineering Limitations**

Limitation	Mechanism	Mitigation
Line-of-sight constraint	Ultrasonic and IR need a clear path; a bicycle, trailer or protruding cargo changes the echo	Magnetometer primary; fusion voting
Environmental noise (outdoor)	Rain, wind, dust, temperature gradients, direct sunlight on IR	Temperature compensation, sealed transducers, 940 nm modulated IR with filters, sensor-health monitoring

Battery degradation	Primary lithium cells lose capacity with low temperature and self-discharge; Li-ion ages by cycle and calendar	Derate capacity 20 to 30 %; report battery voltage; plan replacement cycle
Latency spikes (rural cellular)	Cell congestion, RRC state promotion, retransmissions: RTT from 100 ms to several seconds	Local decision-making, QoS 1 with queue, tolerate eventual consistency
Magnetic cross-talk	Neighbouring vehicles and steel structures	Calibrated thresholds, adaptive baseline, neighbour-aware filtering
Single point of failure	One gateway serves a zone	Redundant gateways; both receive LoRa frames and the cloud de-duplicates by sequence number

10. Real-World Applications

Instructor note: the deployments below are described at a general level; confirm current figures and details from the cited sources before quoting them in a viva.

- **On-street, smart-city (San Francisco SFpark; Barcelona; Santander).** SFpark (SFMTA, 2011 to 2013 pilot) used wireless in-ground sensors with demand-responsive pricing to target a lot-occupancy band, and published evaluation reports [14]. Barcelona and Santander (SmartSantander test-bed) deployed in-ground magnetic sensors in on-street bays with wireless gateways and app/display integration. Key architectural lessons: sealed, buried magnetometer nodes with multi-year batteries and sub-GHz or IEEE 802.15.4 backhaul.
- **Shopping malls and multi-storey garages.** Ceiling-mounted ultrasonic sensors per bay with red/green/blue LED indicators and zone counters on entrance signs, connected over RS-485 or Ethernet to a central server. These emphasise wired reliability and fast visual feedback over battery life. Some also add ANPR at ramps for ticketless billing.
- **Airport multi-level parking.** Large garages use ANPR lanes at entry/exit with per-level counters and guidance, long-stay pre-booking and dynamic pricing tied to flight schedules and demand. They illustrate the value of cloud analytics, payment integration and redundancy.

Related work and surveys of these architectures are summarised in [11] and [12].

11. Challenges and Possible Improvements

11.1 Operational Challenges

- **False triggers.** Pets, shopping carts and debris break an IR beam. Mitigation: vehicle-size gating (ultrasonic distance and echo-duration signature), dual-beam sequencing (the front beam must be interrupted before the rear beam for a valid vehicle), and loop/magnetic confirmation at the gate.
- **Mesh and network scalability.** LoRa star topology scales to hundreds of nodes per gateway but suffers ALOHA collisions: with N nodes each sending a frame of airtime T every period P, the collision probability grows roughly as 1 - exp(-2 N T / P). For N = 300, T = 57 ms and P = 600 s: 1 - exp(-0.057) = about 5.5 %. Mitigations: jittered heartbeat timing, ADR, multiple gateways, or Listen-Before-Talk. BLE or 802.15.4 mesh adds relay energy and routing complexity for little gain at this scale.
- **Cybersecurity.** Threats are spoofed slot data, rogue gate commands, firmware tampering and eavesdropping. Controls: MQTT over TLS 1.2 or 1.3 [9] with server certificate verification and per-device credentials; ACL-restricted topics (a gateway may publish only to its own lot); AES128 for LoRa payloads with frame counters to stop replay; secure boot and signed OTA updates on ESP32 (flash encryption); the gate command channel requires an authenticated and authorised command plus a local interlock (safety beam cannot be bypassed remotely). TLS overhead (RAM about 40 to 60 kB, handshake 1 to 2 s) is acceptable on the mains-powered gateway but not on LoRa nodes, so the security boundary is the gateway.
- **Maintenance.** Sensor fouling and magnetometer drift need self-diagnosis (heartbeat with battery voltage, noise floor, last-valid-echo flag).

11.2 Future Enhancements

- **TinyML edge vision.** An ESP32-CAM (OV2640) running a quantised MobileNet-style classifier with TensorFlow Lite Micro [13]: 96 x 96 greyscale input, int8 weights, roughly 250 kB of model and tens of ms to hundreds of ms inference on a 240 MHz core with PSRAM. Replaces the ultrasonic/IR gate check with vehicle-vs-nonvehicle classification and supports lightweight plate detection; privacy is improved because only the label (not the image) leaves the device.
- **Dynamic pricing.** Price p_t = p_0 x (1 + k (O_t - O_target)) bounded by min/max, where O_t is occupancy (SFpark-style targeting [14]). Requires the forecast pipeline from Section 9.1 and fairness constraints.
- **Solar energy harvesting.** For the gateway-less node or remote lot: a 1 W panel at an average of 4 peak-sun-hours gives about 4 Wh/day; a node using 0.18 mW (4.3 mWh/day) is covered by an area of only a few cm^2, so a small panel plus supercapacitor/LiFePO4 cell eliminates battery replacement. MPPT charging (CN3791 or BQ25570-class) and low-voltage cutoff protection are required.

12. Conclusion and References

12.1 Conclusion

The analysed architecture is technically feasible with commodity components and balances the five objectives by assigning each concern to the layer best suited to it. Battery-powered STM32L4 slot nodes with a magnetometer-primary, ultrasonic-secondary sensing strategy and LoRa uplink reach an estimated 4+ year lifetime (about 60 µA average), roughly 6 x better than an ESP32/Wi-Fi node and 80 x more energy-efficient per message. A mains-powered ESP32 gateway running FreeRTOS isolates the real-time barrier path from network stalls, achieving an ISR-to-actuation target of about 1 ms and endto-end slot visibility of about 1 s. MQTT over TLS gives secure, compact, push-capable transport, with SPIFFS store-and-forward preserving data through outages.

The main trade-offs are accuracy versus cost (magnetometers and cameras versus HC-SR04/IR), energy versus latency (sleep interval versus detection delay: with T = 10 s, worst-case detection is about 10 s, which is why event-driven wake on magnetic interrupt is the next improvement), and security overhead versus node capability. Future work on TinyML, dynamic pricing and energy harvesting would raise accuracy and sustainability, subject to validation by field trials rather than analytical estimates.

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